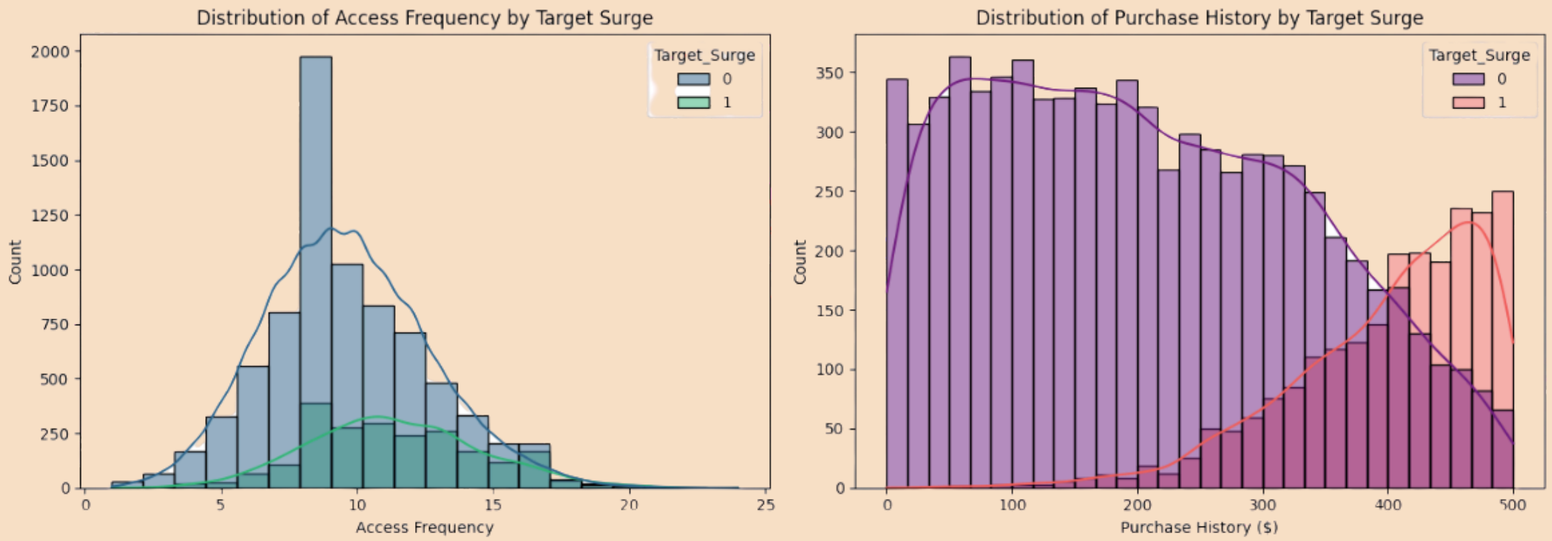
GRAPH REPRESENTATIONS



Pairplot of Features Colored by Target Surge

Image 1 — Feature Distribution by Surge Shows how AccessFrequency and PurchaseHistory behave differently for surge (1) vs non-surge (0) users. Key insight: surge users tend to have higher access frequency and higher purchase history — these are the predictive signals.

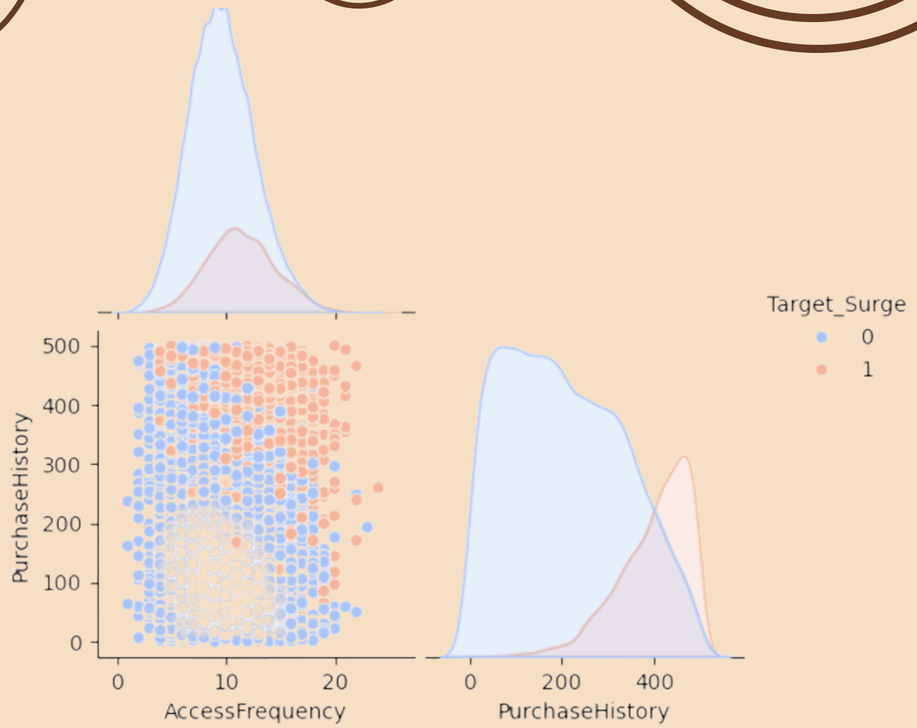


Image 2 — Pairplot of Features A pairplot showing the joint relationship between AccessFrequency and PurchaseHistory, coloured by surge label. The overlap between blue (no surge) and orange (surge) shows why a simple threshold won't work — you need ML to separate them.

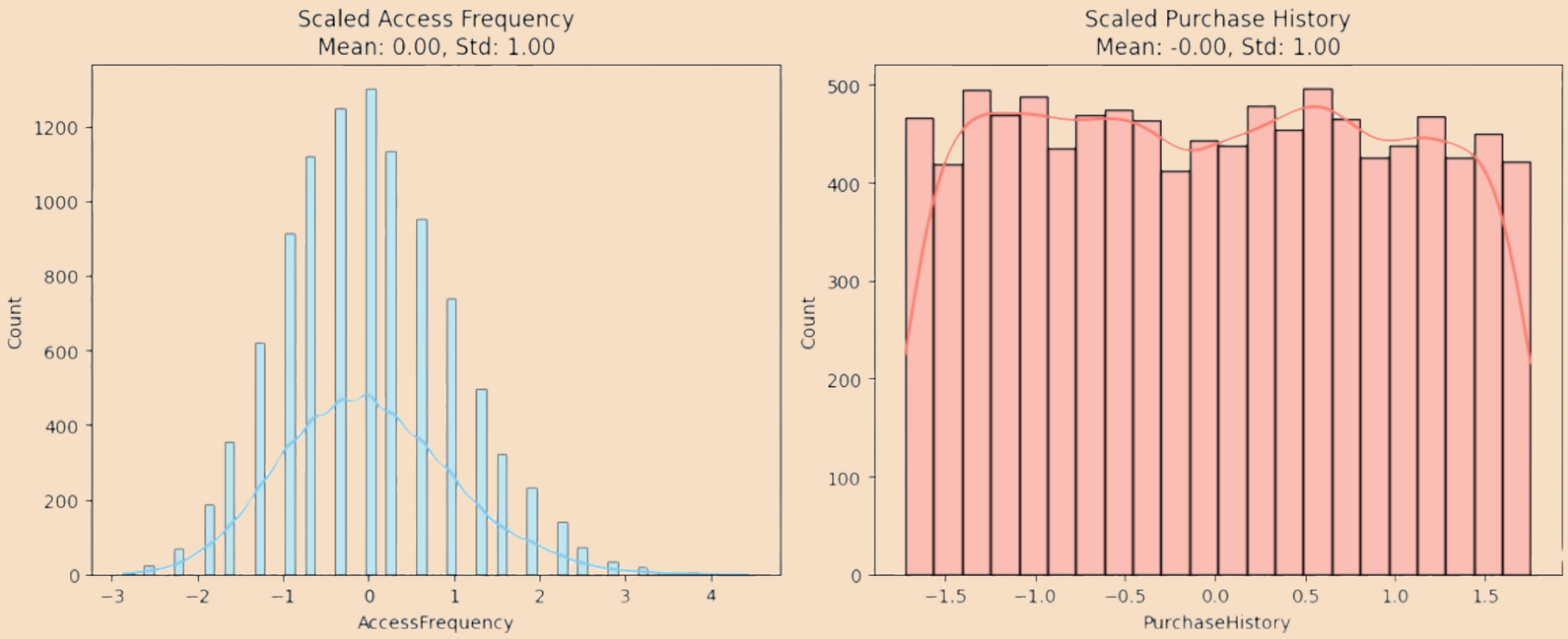


Image 3 — Scaled Feature Distributions Both features have been standardised (mean=0, std=1) using z-score scaling before feeding into the model. AccessFrequency follows a normal-ish bell curve; PurchaseHistory is nearly uniform — meaning purchase history alone is hard to use without clustering.

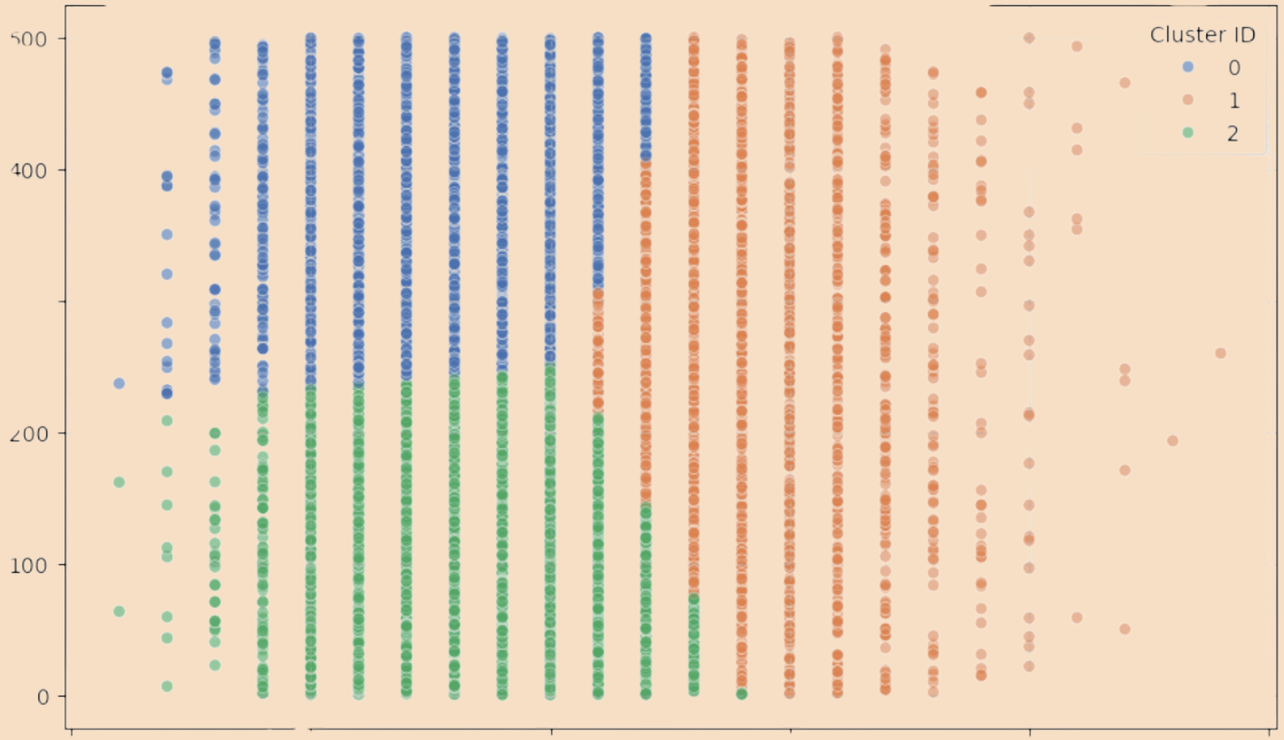
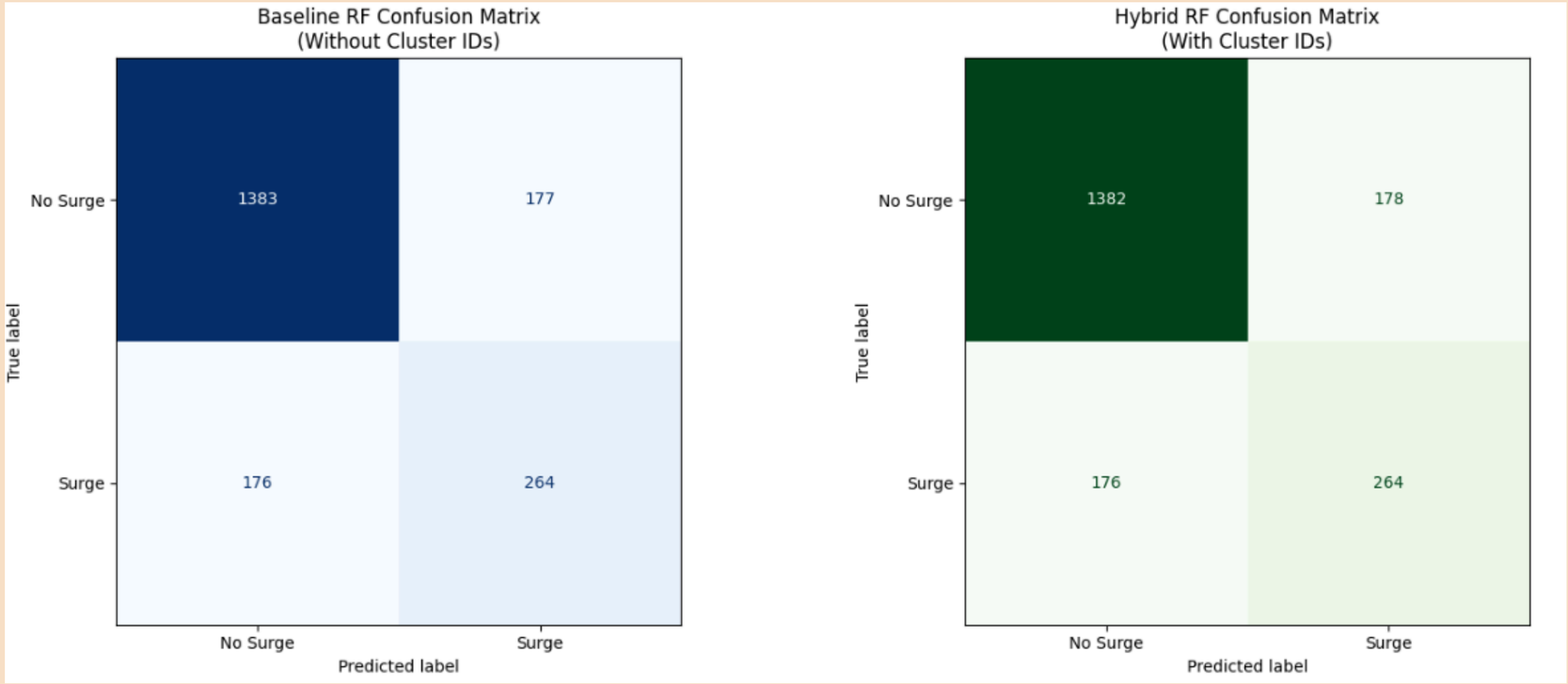


Image 4 — Customer Segmentation (K-Means Clusters) The K-Means clustering result — your 3 behavioural archetypes plotted in feature space. Cluster 0 (blue) = high purchase, low-mid frequency. Cluster 1 (orange) = high frequency, varied spend. Cluster 2 (green) = low purchase, mid frequency. This is Phase 1 of the hybrid model.



- Image 5 — Confusion Matrices: Baseline vs Hybrid RF Compares a standard Random Forest (without cluster info) against the Hybrid RF (with Cluster IDs added as a feature). Both perform similarly here (264 correct surge predictions, 176 missed), suggesting the cluster feature alone didn't dramatically shift raw counts — but precision tells a more nuanced story.

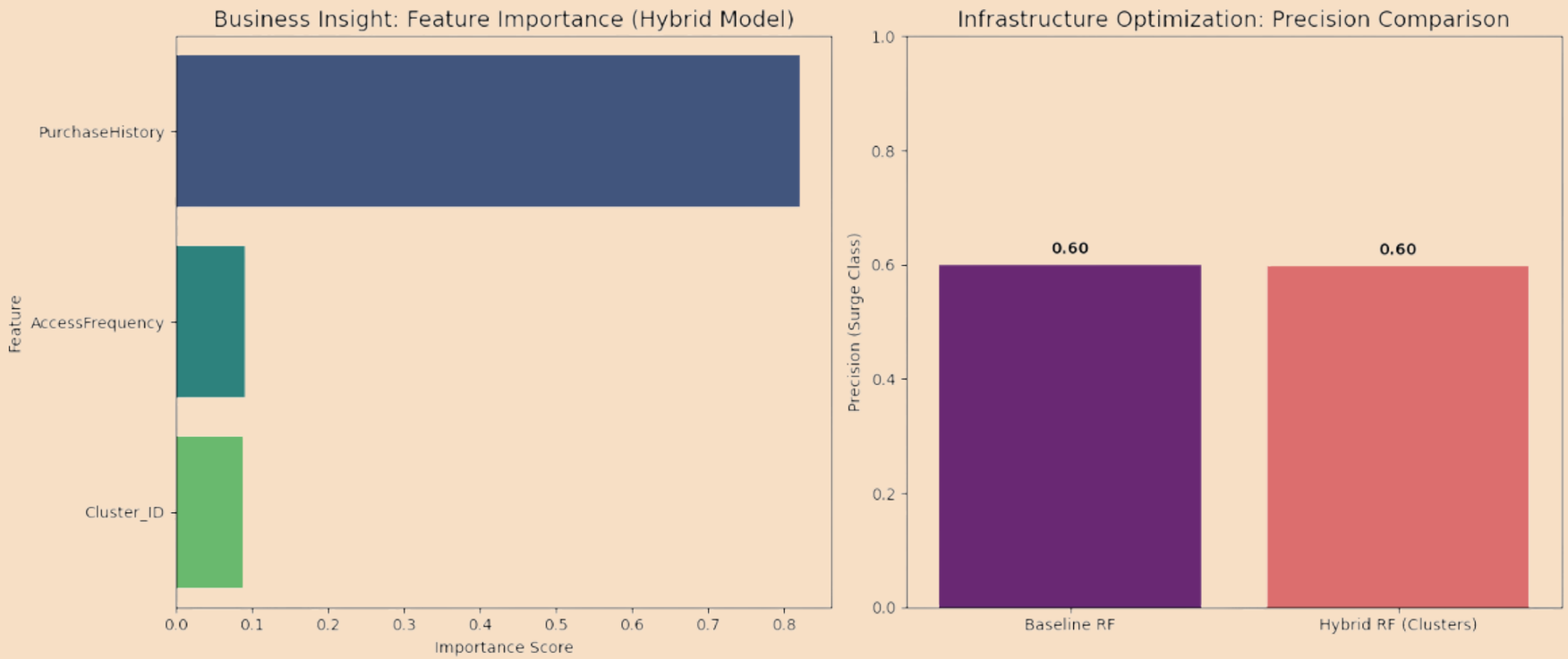


Image 6 — Feature Importance & Precision Comparison Two key business insights: PurchaseHistory dominates feature importance (~0.82), AccessFrequency is second (~0.08), and Cluster_ID contributes minimally (~0.03). Both models hit 60% precision on the surge class — meaning 6 in 10 predicted surges are real, which is meaningful for infrastructure pre-provisioning decisions.

THE BIG PICTURE: THIS IS AN END-TO-END HYBRID ML SYSTEM — CLUSTER USERS BEHAVIOURALLY FIRST, THEN CLASSIFY SURGE RISK — DESIGNED TO HELP CLOUD PROVIDERS ALLOCATE RESOURCES BEFORE DEMAND SPIKES, REDUCING BOTH OVER-PROVISIONING COSTS AND SERVICE FAILURES.